# A proof exposition of static Lamé uniqueness in three dimensions

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12 October 2026

**Abstract**

Can static boundary displacement and traction determine both spatially varying elastic moduli? This note presents OpenAI's uniqueness argument for arbitrary smooth isotropic Lamé coefficients with positive elastic energy on any bounded connected smooth domain in three dimensions. Equal boundary maps allow exact physical solutions to be matched in the exterior. Their leading amplitudes then give a comparison that varies holomorphically with the complex direction; the polarization constraint forces this comparison to be the identity and determines both coefficients. We give the complete argument, including a direct local energy proof of the scaled elliptic estimate needed to control the physical corrections.

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## 1 The inverse problem and its uniqueness theorem

The static isotropic elasticity equation relates a displacement field to the stress generated by its strain. Its two Lamé coefficients encode the shear and bulk responses. The inverse boundary problem

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asks whether imposing all boundary displacements and measuring the resulting tractions determines both coefficient functions in the interior. The data are at zero frequency and the coordinates are fixed Euclidean coordinates.

The theorem and main proof mechanism presented here are due to OpenAI [1]. This note gives an attributed exposition of that argument, with additional details for its analytic estimates. Boundary determination fixes the coefficient jets, but interior recovery requires a further comparison of solutions. The essential point is to preserve the physical displacement together with its actual divergence throughout that comparison. Their leading terms occupy a constrained three-dimensional space. Its dependence on the complex direction supplies the rigidity that recovers both moduli.

We first state the data and result. Section 2 turns equality of boundary maps into exterior matching. Sections 3–4 construct the physical solutions used in that matching. The comparison and coefficient recovery are proved in Sections 5 and 6. Appendix A supplies the analytic preliminaries, including a direct quantitative elastic estimate.

Let $\Omega \subset \mathbb{R}^3$ be a bounded connected domain with smooth boundary. For real functions $\lambda, \mu \in C^\infty(\overline{\Omega})$ satisfying

$$\mu > 0, \qquad 3\lambda + 2\mu > 0 \quad \text{on } \overline{\Omega}, \tag{1.1}$$

define the strain, stress, and static elasticity operator by

$$e(u) = \frac{\nabla u + (\nabla u)^T}{2}, \qquad \sigma_{\lambda,\mu}(u) = \lambda(\operatorname{div} u)I + 2\mu e(u), \qquad L_{\lambda,\mu}u = \operatorname{div}\sigma_{\lambda,\mu}(u). \tag{1.2}$$

Here $\mu$ is the shear modulus and $\lambda + 2\mu/3$ is the bulk modulus. The inequalities in (1.1) give uniformly positive elastic energy. For each $f \in H^{1/2}(\partial\Omega;\mathbb{R}^3)$ there is a unique $u_f \in H^1(\Omega;\mathbb{R}^3)$ with $L_{\lambda,\mu}u_f = 0$ and trace $f$. The displacement-to-traction map is defined weakly by

$$\langle \Lambda_{\lambda,\mu} f, g\rangle = \int_\Omega [\lambda(\operatorname{div} u_f)(\operatorname{div} v_g) + 2\mu e(u_f) : e(v_g)]\,\mathrm{d}x, \tag{1.3}$$

where $v_g \in H^1(\Omega;\mathbb{R}^3)$ has trace $g$, and $A : B = \operatorname{tr}(A^T B)$. The weak equation makes this expression independent of the extension. For smooth boundary data it is the traction $\sigma_{\lambda,\mu}(u_f)n$, with $n$ the outward normal.

**Theorem 1.1** (OpenAI)**.** *Let $\Omega \subset \mathbb{R}^3$ be any bounded connected domain with $C^\infty$ boundary. For $j = 1, 2$, let $\lambda_j, \mu_j \in C^\infty(\overline{\Omega};\mathbb{R})$ satisfy $\mu_j > 0$ and $3\lambda_j + 2\mu_j > 0$ on $\overline{\Omega}$. If*

$$\Lambda_{\lambda_1,\mu_1} = \Lambda_{\lambda_2,\mu_2} \colon H^{1/2}(\partial\Omega;\mathbb{R}^3) \longrightarrow H^{-1/2}(\partial\Omega;\mathbb{R}^3),$$

*then $\lambda_1 = \lambda_2$ and $\mu_1 = \mu_2$ throughout $\Omega$.*

Theorem 1.1, reconstructed from OpenAI [1], gives a positive answer to the smooth three-dimensional isotropic elastic Calderón uniqueness problem under (1.1). It uses the full zero-frequency boundary operator in fixed Euclidean coordinates. The coefficients need not be analytic or close to constants, and their agreement near the boundary is not an additional hypothesis.

**Conventions and analytic background.** All coefficients are real. Differential identities and boundary maps are complexified by linearity. Dot products are complex bilinear; Hilbert inner products are Hermitian and linear in the first variable. Measures are Lebesgue measures. Our Fourier transform is $\widehat{v}(\xi) = (2\pi)^{-d/2}\int e^{-ix\cdot\xi}v(x)\,\mathrm{d}x$. The inverse has the opposite exponential sign. Define $H^s(\mathbb{R}^d)$ by the weight $(1+|\xi|^2)^{s/2}$, local spaces by cutoffs, and spaces on open sets by restriction (integer nonnegative norms are equivalent to the derivative norms). Distributions

act on compactly supported smooth tests. Equalities of Sobolev elements and weak limits are understood almost everywhere; smooth coefficient equalities in the conclusion are pointwise. $C^\infty(\overline{\Omega})$ denotes smoothness up to the boundary; the collar construction in Appendix A also supplies a neighborhood extension. All asymptotic estimates below fix the coefficients, balls, local frame and nonzero null direction before sending $\tau \to \infty$. No estimate is claimed uniform over all directions. Appendix A proves the functional, planar and elliptic bridges used below. Its only Fourier input is the explicitly normalized version of Salo's theorems [5]. The three principal cited inputs are boundary jets (Tan–Liu), a scalar Carleman estimate (Salo–Tzou), and the Oka principle for disk frames (Forstnerič). Their hypotheses and roles are given at the points where they are used.

## 2 Boundary reduction and physical transfer

Throughout the proof, the two coefficient pairs satisfy the hypotheses of Theorem 1.1 and have equal displacement-to-traction maps.

**Lemma 2.1** (Common exterior extension)**.** *The two pairs admit real smooth extensions to $\mathbb{R}^3$ that satisfy (1.1) everywhere, coincide on $\mathbb{R}^3 \setminus \Omega$, and equal a common constant admissible pair outside a ball.*

*Proof.* By Theorem 1.2 of Tan and Liu [2], equality of the elastic Dirichlet-to-Neumann maps determines all boundary partial derivatives of both Lamé moduli. Our inequalities imply $\lambda + \mu > \mu/3 > 0$, so their positivity condition holds. Their definition of the boundary operator is $\lambda(\operatorname{div} u)n + \mu(\nabla u + (\nabla u)^T)n$, so its Euclidean specialization agrees with (1.3). Consequently the two pairs have identical jets at every point of $\partial\Omega$.

Smoothly extend the first pair to a neighborhood of $\overline{\Omega}$. By compactness and strict positivity, shrink the neighborhood so that (1.1) still holds there. Choose a smooth cutoff equal to one on a smaller neighborhood of $\overline{\Omega}$ and supported in the first neighborhood. Interpolate with any constant pair satisfying (1.1). The admissible set in the $(\lambda, \mu)$ plane is convex, so this produces a smooth admissible global extension of the first pair, constant outside a compact set.

For the second pair use its prescribed values in $\Omega$ and the first extension on the complement. Equality of all jets makes this piecewise definition smooth across each boundary component. Positivity holds on both sides and on the boundary. This proves the assertion, including when the complement of $\Omega$ has bounded components. □

Fix these extensions and retain the notation $\lambda_j, \mu_j$ and $L_j = L_{\lambda_j,\mu_j}$ for them. Choose $r > 0$ so that $\overline{\Omega} \subset B_r$ and the extensions are constant outside $B_r$. Here $B_R$ is the open ball of radius $R$ centered at the origin.

**Proposition 2.2** (Transfer of physical solutions)**.** *Let $B$ be a ball containing $\overline{B}_r$. Every smooth solution $u_1$ of $L_1 u_1 = 0$ on $B$ has a smooth counterpart $u_2$ satisfying $L_2 u_2 = 0$ on $B$ and*

$$u_2 = u_1 \quad \text{on } B \setminus \overline{\Omega}.$$

*In particular, $u_2 - u_1$ and $\operatorname{div} u_2 - \operatorname{div} u_1$ are smooth and supported in $\overline{\Omega}$.*

*Proof.* Solve the second Dirichlet problem in $\Omega$ with trace $u_1|_{\partial\Omega}$, and use $u_1$ outside $\overline{\Omega}$. Matching traces give an $H^1_{\mathrm{loc}}(B)$ field $u_2$. The restrictions of the two interior solutions have identical boundary tractions because their Dirichlet values and boundary maps agree. For a smooth test vector compactly supported in $B$, the weak boundary term contributed by the interior therefore agrees

with that for $u_1$. The coefficients and displacements coincide on the exterior, where the same test contributes the opposite interface term. Hence the glued field satisfies $L_2 u_2 = 0$ weakly on $B$.

The extended operator is smooth and strongly elliptic. Lemma A.4, applied also across $\partial\Omega$, gives $u_2 \in C^\infty(B;\mathbb{C}^3)$. The difference is zero off $\overline{\Omega}$, as are all its derivatives there. Its support and the support of its divergence are thus contained in $\overline{\Omega} \Subset B_r$. □

## 3 An augmented system and analytic estimates

This section constructs a scalar-Laplacian augmented operator and a supported estimate that will control differences of transferred physical displacement–divergence pairs. The identities below give the formulation used in this proof. We first work with one of the smoothly extended coefficient pairs from Lemma 2.1, and suppress its subscript. Set

$$m = \mu, \qquad k = \lambda + \mu, \qquad \ell = \lambda + 2\mu = k + m.$$

The energy assumptions imply $m > 0$, $k > m/3$, and $\ell > 4m/3$. For an independent vector–scalar pair $U = (u, d)$, define

$$\begin{aligned} R(u,d)_i &= m\Delta u_i + k\partial_i d + \sum_{j=1}^{3} (\partial_j m)(\partial_j u_i + \partial_i u_j) + (\partial_i \lambda) d, \\ S(u,d) &= \ell \Delta d + 2\nabla k \cdot \nabla d + 2\nabla m \cdot \Delta u + 2 \sum_{i,j=1}^{3} (\partial_i \partial_j m)\partial_i u_j + (\Delta\lambda) d, \\ P &= m\Delta + \nabla m \cdot \nabla. \end{aligned} \tag{3.1}$$

The scalar $d$ will equal $\operatorname{div} u$ for physical solutions, but keeping it independent gives the compatibility identity needed below.

**Lemma 3.1.** *For every smooth pair $(u, d)$,*

$$Lu = R(u, \operatorname{div} u), \qquad \operatorname{div} R(u,d) - S(u,d) = P(\operatorname{div} u - d). \tag{3.2}$$

*In particular, $Lu = 0$ implies $R(u, \operatorname{div} u) = S(u, \operatorname{div} u) = 0$. The normalized augmented operator*

$$\mathcal{L}U = \Big( m^{-1} R(u,d),\ \ell^{-1}\big(S(u,d) - 2m^{-1}\nabla m \cdot R(u,d)\big)\Big) \tag{3.3}$$

*has the form*

$$\mathcal{L} = \Delta \mathrm{Id}_4 + \sum_{j=1}^{3} A_j(x)\partial_j + A_0(x) \tag{3.4}$$

*with smooth matrix coefficients. Consequently, the difference of the normalized augmented operators for the two coefficient pairs has order at most one and coefficients supported in $\overline{\Omega}$.*

*Proof.* The first identity is the coordinate expansion of $\operatorname{div} \sigma(u)$. To verify the second independently of the constraint, write $v = \operatorname{div} u$. Differentiating (3.1) gives

$$\begin{aligned} \operatorname{div} R(u,d) = {} & m\Delta v + \nabla m \cdot \nabla v + k\Delta d + (\nabla k + \nabla\lambda)\cdot\nabla d \\ & + 2\nabla m \cdot \Delta u + 2 \sum_{i,j=1}^{3} (\partial_i\partial_j m)\partial_i u_j + (\Delta\lambda) d. \end{aligned}$$

Subtracting $S$ and using $k = \lambda + m$ gives $m\Delta(v-d) + \nabla m \cdot \nabla(v-d)$.

For the principal part in (3.3), the only second derivatives of $u$ in its scalar numerator cancel. More explicitly, if

$$C_m(u)_i = \sum_{j=1}^{3} (\partial_j m)(\partial_j u_i + \partial_i u_j),$$

then this numerator is

$$\begin{aligned} S - 2m^{-1}\nabla m \cdot R = \ell\Delta d + (2\nabla k - 2km^{-1}\nabla m)\cdot\nabla d \\ + 2\sum_{i,j=1}^{3} (\partial_i\partial_j m)\partial_i u_j - 2m^{-1}\nabla m \cdot C_m(u) \\ + (\Delta\lambda - 2m^{-1}\nabla m\cdot\nabla\lambda)d. \end{aligned}$$

This proves (3.4). The support assertion follows from equality of the extended coefficients off $\overline{\Omega}$. □

The order-one difference of the two augmented operators is the forcing term to be estimated after physical transfer in Section 5.

Fix a nonzero $\theta \in \mathbb{C}^3$ with $\theta \cdot \theta = 0$, and write

$$D_\theta = \theta \cdot \nabla, \qquad E_\tau(x) = e^{\tau\theta\cdot x}, \qquad h = \tau^{-1}. \tag{3.5}$$

For $s \in \mathbb{R}$, the semiclassical Sobolev norm on $\mathbb{R}^3$ is

$$\|v\|_{H_h^s}^2 = \int_{\mathbb{R}^3} (1 + h^2\,|\xi|^2)^s\,|\widehat{v}(\xi)|^2\,\mathrm{d}\xi.$$

For vectors, we sum the component norms. These norms and all Hilbert space adjoints below use the Hermitian structure, whereas products such as $\theta\cdot\theta$ remain complex bilinear. Constants in the following estimates may depend on the coefficients, the fixed ball, and $\theta$, but not on sufficiently large $\tau$.

**Proposition 3.2** (Supported Carleman estimate)**.** *Let $B$ be a bounded ball. For $U \in C_c^\infty(B;\mathbb{C}^4)$ and all sufficiently small $h > 0$,*

$$\left\|E_\tau^{-1}U\right\|_{H_h^1} \le Ch\left\|E_\tau^{-1}\mathcal{L}U\right\|_{H_h^{-1}}. \tag{3.6}$$

*Proof.* Put $\varphi(x) = -\operatorname{Re}(\theta\cdot x)$. A nonzero complex null vector has nonzero real part, so $\varphi$ is a nonconstant linear limiting Carleman weight for the Laplacian. For $\varphi_\varepsilon = \varphi + h\varphi^2/(2\varepsilon)$, [3, Lemma 2.1], with Sobolev index $s = -1$, gives

$$\frac{h}{\sqrt{\varepsilon}}\,\|v\|_{H_h^1} \le C\left\|e^{\varphi_\varepsilon/h}h^2\Delta e^{-\varphi_\varepsilon/h}v\right\|_{H_h^{-1}}, \qquad v \in C_c^\infty(B), \tag{3.7}$$

for $h \ll \varepsilon \ll 1$, with $C$ independent of sufficiently small $\varepsilon$. Its statement has a gain of two derivatives and permits every real $s$. The sign of the Laplacian is immaterial here.

Apply (3.7) componentwise. A first-order term in (3.4) contributes

$$e^{\varphi_\varepsilon/h}h^2A_j\partial_j e^{-\varphi_\varepsilon/h}v = hA_j(h\partial_j v) - hA_j(\partial_j\varphi_\varepsilon)v.$$

Its $H_h^{-1}$ norm is bounded by $C_1 h \|v\|_{H_h^1}$, uniformly when $h/\varepsilon$ is small. The zeroth-order terms have the same bound. Choose $\varepsilon > 0$ so small that these terms are absorbed by the left side of (3.7), and then choose $h$ sufficiently small relative to this fixed $\varepsilon$. We obtain

$$\left\| e^{\varphi_\varepsilon/h} U \right\|_{H_h^1} \le C_\varepsilon h \left\| e^{\varphi_\varepsilon/h} \mathcal{L} U \right\|_{H_h^{-1}}.$$

The factor $e^{\varphi^2/(2\varepsilon)}$ is now fixed. It and its inverse, smoothly localized around $\overline{B}$, are bounded multipliers on $H_h^1$, uniformly for $0 < h \le 1$, by the product rule. They are bounded on $H_h^{-1}$ by duality. Removing this factor gives the same estimate with $\varphi$ in place of $\varphi_\varepsilon$. The multiplier constants may depend on $\varepsilon$; no subsequent absorption involves them.

Finally, write $\beta = \operatorname{Im}\theta$. Multiplication by $e^{-i\beta\cdot x/h}$ shifts the semiclassical frequency $h\xi$ by the fixed vector $-\beta$. The weights $1 + |h\xi|^2$ and $1 + |h\xi - \beta|^2$ are comparable, so this multiplier and its inverse are uniformly bounded on $H_h^{\pm 1}$. Since $E_\tau^{-1} = e^{\varphi/h} e^{-i\beta\cdot x/h}$, this proves (3.6). □

**Corollary 3.3.** *For the same $B$ and $\theta$, and all sufficiently large $\tau$, every $w \in C_c^\infty(B;\mathbb{C}^3)$ satisfies*

$$\left\| E_\tau^{-1} w \right\|_{L^2(B)} \le C \left\| E_\tau^{-1} L w \right\|_{L^2(B)}. \tag{3.8}$$

*Moreover, for every $f \in L^2(B;\mathbb{C}^3)$ there exists $z \in L^2(B;\mathbb{C}^3)$ satisfying, distributionally,*

$$E_\tau^{-1} L (E_\tau z) = f \quad \text{in } B, \qquad \|z\|_{L^2(B)} \le C \|f\|_{L^2(B)}. \tag{3.9}$$

*Proof.* Apply Proposition 3.2 to $(w, \operatorname{div} w)$. By (3.2), its augmented forcing is formed from $Lw$ and $\operatorname{div}(Lw)$. If $F = E_\tau^{-1} L w$, then

$$E_\tau^{-1} \operatorname{div}(Lw) = \operatorname{div} F + \tau\theta \cdot F.$$

The Fourier definition of the norms gives $\|\partial_j F\|_{H_h^{-1}} \le h^{-1} \|F\|_{L^2}$. Smooth coefficient multipliers, localized near $\overline{B}$, are uniformly bounded on $H_h^{-1}$. Hence

$$\left\| E_\tau^{-1} \mathcal{L}(w, \operatorname{div} w) \right\|_{H_h^{-1}} \le C h^{-1} \|F\|_{L^2}.$$

Taking the displacement component on the left side of (3.6) proves (3.8).

Let $A_\theta = E_\tau^{-1} L E_\tau$ on test functions in $B$. Substituting $w = E_\tau v$ in (3.8) gives $\|v\|_{L^2} \le C \|A_\theta v\|_{L^2}$. Formal self-adjointness of $L$ gives

$$A_\theta^* = e^{\tau\overline{\theta}\cdot x} L e^{-\tau\overline{\theta}\cdot x} = A_{-\overline{\theta}}.$$

The same estimate for the null direction $-\overline{\theta}$ thus holds for $A_\theta^*$. Use the $L^2$ inner product linear in its first argument. The functional

$$A_\theta^* v \longmapsto (v, f)_{L^2(B)}, \qquad v \in C_c^\infty(B;\mathbb{C}^3),$$

is well-defined and has norm at most $C\|f\|_{L^2}$. Extend it continuously to the closure of this range, a closed Hilbert subspace, and use the Hilbert representation proved in Appendix A to represent it as $(\,\cdot\,, z)_{L^2}$. Then $(A_\theta^* v, z) = (v, f)$ for every test function $v$, which is the distributional equation in (3.9). The norm of the representing vector gives the asserted bound. □

**Lemma 3.4** (Scaled interior estimate)**.** *Let $B' \Subset B$. There exists $h_{B'} > 0$ such that, whenever $0 < h = \tau^{-1} < h_{B'}$ and $z, f \in L^2(B;\mathbb{C}^3)$ satisfy $E_\tau^{-1} L(E_\tau z) = f$ distributionally in $B$,*

$$\sum_{j=0}^{2} h^j \left\| \nabla^j z \right\|_{L^2(B')} \le C_{B'} \left( \|z\|_{L^2(B)} + h^2 \|f\|_{L^2(B)} \right). \tag{3.10}$$

*If $f$ is smooth, then $z$ is smooth in $B$ for each fixed $\tau$.*

*Proof.* Use Lemma A.5 of Appendix A. For $x = x_0 + hy$, multiply the conjugated equation by $-h^2$. Its principal part is the negative divergence of the real elastic stiffness tensor at $x_0 + hy$. Its remaining coefficients, including all first derivatives, are bounded uniformly for $x_0 \in \overline{B'}$ and $0 < h \le h_0$; the factors $h^{-1}, h^{-2}$ from conjugation are canceled by the scaling. Strict energy positivity on $\overline{B}$ gives a uniform coercivity constant. The lemma therefore gives

$$\sum_{j=0}^{2} h^j \|\nabla^j z\|_{L^2(B(x_0,h))} \le C\big( \|z\|_{L^2(B(x_0,2h))} + h^2 \|f\|_{L^2(B(x_0,2h))} \big).$$

Choose $h$ small enough that every doubled ball is in $B$. A grid of mesh comparable to $h$ covers $B'$ with these inner balls; each point belongs to at most a dimension-dependent number of doubled balls. Square and sum the estimates, then take square roots, to get the asserted bound. The qualitative part of Lemma A.4, for fixed $h$, gives smoothness when $f$ is smooth. The constants may depend on the fixed direction, coefficients and $\operatorname{dist}(B', \partial B)$. □

## 4 Physical solutions realizing local transport frames

We now construct exact elasticity solutions with controlled leading behavior for both the displacement and its actual divergence. We begin by identifying the equations obeyed by these leading terms. Fix the null vector $\theta$ from (3.5), and write $D = D_\theta$ when convenient. Since $\theta \cdot \theta = 0$, conjugating the operators in (3.1) gives

$$E_\tau^{-1} R(E_\tau a, E_\tau b) = R(a,b) + \tau R^\theta(a,b), \qquad E_\tau^{-1} S(E_\tau a, E_\tau b) = S(a,b) + \tau S^\theta(a,b),$$

where

$$\begin{aligned} R^\theta(a,b) &= 2mDa + (Dm)a + \theta(kb + \nabla m \cdot a), \\ S^\theta(a,b) &= 2\ell Db + 2(Dk)b + 4\nabla m \cdot Da + 2(D\nabla m) \cdot a. \end{aligned} \tag{4.1}$$

For smooth vector amplitudes $a_0, a_1$, the first two terms of a trial displacement give

$$E_\tau^{-1} \operatorname{div}\big(E_\tau(a_0 + \tau^{-1} a_1)\big) = \tau\theta \cdot a_0 + \big(\operatorname{div} a_0 + \theta \cdot a_1\big) + \tau^{-1} \operatorname{div} a_1.$$

An $O(1)$ leading divergence therefore requires $\theta \cdot a_0 = 0$, while its scalar term also contains $\theta \cdot a_1$ and need not equal $\operatorname{div} a_0$. We retain that scalar as the component $b$ of the leading pair. Define the fixed complex vector spaces

$$H_\theta = \{a \in \mathbb{C}^3 : \theta \cdot a = 0\}, \qquad \mathcal{E}_\theta = H_\theta \oplus \mathbb{C}. \tag{4.2}$$

The transport equation on $\mathcal{E}_\theta$ is

$$R^\theta(a,b) = 0, \qquad S^\theta(a,b) = 0, \qquad (a,b) \in \mathcal{E}_\theta. \tag{4.3}$$

Indeed, putting $P^\theta = 2mD + (Dm)$, we have

$$\theta \cdot R^\theta(a,b) = P^\theta(\theta \cdot a). \tag{4.4}$$

Thus $R^\theta$ takes values in $H_\theta$ when $a$ does. On $\mathcal{E}_\theta$, the coefficient multiplying $D(a,b)$ in the two transport equations is

$$\begin{pmatrix} 2m\mathrm{Id}_{H_\theta} & 0 \\ 4\nabla m\cdot & 2\ell \end{pmatrix}.$$

It is invertible, so (4.3) is a square rank-three system with principal part $D$.

For clarity, write $\theta = \alpha + i\beta$ with $\alpha, \beta \in \mathbb{R}^3$. The null condition says $|\alpha| = |\beta| = \rho > 0$ and $\alpha \cdot \beta = 0$. Let $(y_1, y_2, t)$ be positively oriented orthonormal coordinates in the directions $\alpha/\rho$, $\beta/\rho$, and their cross product. Then, for $z = y_1 + iy_2$,

$$D = \rho(\partial_{y_1} + i\partial_{y_2}) = 2\rho\partial_{\bar z}.$$

We denote the planar disk $\{y \in \mathbb{R}^2 : |y| < R\}$ by $D^2_R$.

For each chosen transverse coordinate $t_0$, we will select a smooth frame of (4.3) for $t$ near $t_0$ and then realize its three columns by physical solutions. The next lemma records the local parameter dependence and inhomogeneous solvability with transverse support needed here.

**Lemma 4.1** (Planar frames and inhomogeneous transports)**.** *Let $I \subset \mathbb{R}$ be an open interval and let $A \in C^\infty(D^2_R \times I; \mathbb{C}^{n\times n})$. For each fixed $t \in I$, the equation*

$$\partial_{\bar z} V = A(z,t)V \tag{4.5}$$

*has a smooth invertible matrix solution on $D^2_R$. For every $R' < R$ and $t_0 \in I$, there are an interval $I_0 \Subset I$ containing $t_0$ and a smooth invertible solution $V$ on $D^2_{R'} \times I_0$.*

*Given this $V$, a radius $R'' < R'$, and $F \in C^\infty(D^2_{R'} \times I_0; \mathbb{C}^n)$, the equation*

$$\partial_{\bar z} v = Av + F \tag{4.6}$$

*has a smooth solution on $D^2_{R''} \times I_0$. If $F(\,\cdot\,, t) = 0$ for $t \notin K$, where $K \Subset I_0$ is compact, the solution can be chosen with the same transverse support property. These assertions also hold with $D$ in place of $\partial_{\bar z}$, and for systems obtained by multiplying $D$ by an invertible smooth matrix.*

*Proof.* We first construct frames locally at a fixed parameter. For a bounded function $f$ supported in a disk of radius $\delta$, the Cauchy transform

$$(\mathcal{T}f)(z) = \frac{1}{\pi}\int_{\mathbb{C}} \frac{f(w)}{z-w}\,\mathrm{d}y_1(w)\,\mathrm{d}y_2(w)$$

satisfies $\partial_{\bar z}\mathcal{T}f = f$ in distributions and $\|\mathcal{T}f\|_{L^\infty} \le C\delta\,\|f\|_{L^\infty}$. The latter estimate follows by integrating $|z-w|^{-1}$ over the support disk. Choose a smooth cutoff $\chi$ supported in a sufficiently small disk and equal to one on a smaller disk. On bounded matrix functions, the equation

$$V = \mathrm{Id}_n + \mathcal{T}(\chi A V)$$

is then solved by a convergent Neumann series. Shrinking the disk until the operator norm is less than $1/3$ gives $\|V - \mathrm{Id}_n\|_{L^\infty} < 1/2$, hence invertibility. The distributional equation and Lemma A.4 for $\partial_{\bar z}$ show that $V$ is smooth on the smaller disk. This proves local smooth invertible solvability.

If $V_i, V_j$ are two such local frames, then $\partial_{\bar z}(V_i^{-1}V_j) = 0$ on their overlap. Their transition functions therefore define a holomorphic vector bundle on $D^2_R$, whose underlying smooth bundle

is trivial. Its holomorphic frame bundle has fiber $\mathrm{GL}_n(\mathbb{C})$. The disk is Stein and contractible, so a continuous section of this frame bundle exists. The Oka principle for sections of holomorphic fiber bundles with complex homogeneous fibers, in the form of [4, Theorem 1.1, p. 1018], supplies a holomorphic section. In the original smooth trivialization it is a smooth invertible matrix solving (4.5) throughout $D^2_R$.

To obtain local parameter dependence, choose such a frame $V_0(z)$ at $t_0$. After writing $V = V_0 W$, the equation becomes

$$\partial_{\bar z} W = B(z,t)W, \qquad B(z,t) = V_0(z)^{-1}(A(z,t) - A(z,t_0))V_0(z).$$

Choose a cutoff $\chi \in C^\infty_c(D^2_R)$ equal to one near $\overline{D}^2_{R'}$. On its compact support, $B(\,\cdot\,,t)$ tends uniformly to zero as $t \to t_0$. On a sufficiently small interval $I_0 \Subset I$, solve

$$W = \mathrm{Id}_n + \mathcal{T}(\chi B(\,\cdot\,,t)W)$$

by a Neumann series with norm less than $1/3$. This gives an invertible solution on $D^2_{R'} \times I_0$. The integral operator depends smoothly on $t$ in operator norm on $L^\infty$; its inverse does also. Hence all parameter derivatives of $W$ exist in $L^\infty$. Differentiating its equation and applying interior elliptic regularity successively gives smoothness jointly in $(z,t)$, with bounds on compact subsets. Thus $V = V_0 W$ has the claimed parameter regularity.

Finally, choose $\chi \in C^\infty_c(D^2_{R'})$ equal to one near $\overline{D}^2_{R''}$. The formula

$$v = V\mathcal{T}(\chi V^{-1} F)$$

solves (4.6) on the smaller disk. The transform acts only in the planar variable, so it preserves the specified support in $t$. Dividing a system with invertible derivative coefficient by that coefficient, and using $D = 2\rho\partial_{\bar z}$, proves the final assertion. □

To carry a leading frame through physical transfer, we need control of both the displacement–divergence pair and its derivatives. Corollary 3.3 gives physical solvability with an $L^2$ estimate independent of $\tau$. We will make the correction small enough after taking a divergence by first constructing a compatible expansion of truncation order $N \geq 4$. The proof below obtains a physical residual of size $O(\tau^{1-N})$ and a conjugated divergence correction of size $O(\tau^{2-N})$. The resulting strong convergence identifies the first medium's leading columns and, by exterior agreement, the exterior values of the transferred columns. The weighted $O(\tau)$ derivative bound controls the first-order forcing obtained by applying the difference of the two normalized augmented operators to the first physical pair.

**Proposition 4.2** (Physical realization of local transport frames)**.** *For every nonzero null vector $\theta$ and every $t_0 \in (-2r, 2r)$, there exist an open interval $J \subset (-2r, 2r)$ containing $t_0$ and three smooth solutions $(a_0^\nu, b_0^\nu)$, $1 \leq \nu \leq 3$, of (4.3) on a neighborhood of $\overline{B}_{5r}$, with the following properties.*

*The three columns form a frame of $\mathcal{E}_\theta$ at every point of $D^2_{2r} \times J$. For each $\nu$ and all sufficiently large $\tau$, there is $u^\nu_\tau \in C^\infty(B_{5r}; \mathbb{C}^3)$ with $Lu^\nu_\tau = 0$ such that, on putting $U^\nu_\tau = (u^\nu_\tau, \operatorname{div} u^\nu_\tau)$,*

$$\begin{aligned} E_\tau^{-1} U^\nu_\tau &\longrightarrow (a_0^\nu, b_0^\nu) \quad \text{in } L^2(B_{4r}), \\ \left\| E_\tau^{-1} \nabla U^\nu_\tau \right\|_{L^2(B_{4r})} &= O(\tau). \end{aligned} \tag{4.7}$$

*Proof.* We construct compatible formal amplitudes and then correct the truncated displacement using the physical operator. Fix an integer $N \geq 4$ and radii $R_0 > R_1 > \cdots > R_N > 5r$. The

coefficients are smooth on all of $\mathbb{R}^3$. Apply Lemma 4.1 to the rank-three system (4.3), starting on a disk strictly larger than $D^2_{R_0}$. It supplies a smooth invertible frame

$$\mathcal{F}(y,t) : \mathbb{C}^3 \longrightarrow \mathcal{E}_\theta \quad \text{on } D^2_{R_0} \times I_0,$$

where $I_0 \Subset (-2r, 2r)$ contains $t_0$. Choose an interval $J$ containing $t_0$ with $\overline{J} \subset I_0$, and choose $\chi \in C_c^\infty(I_0)$ equal to one on $J$. Multiply each column of $\mathcal{F}$ by $\chi(t)$ and extend it by zero in $t$ to obtain $(a_0^\nu, b_0^\nu)$ on $D^2_{R_0} \times \mathbb{R}$. Since $D\chi = 0$, these columns still solve (4.3). They are a frame on $D^2_{2r} \times J$, and all have transverse support in the fixed compact set $K = \operatorname{supp}\chi \Subset I_0$.

We construct a physical solution for any one of these columns and suppress $\nu$. The recursion here must cancel the vector residual while making the conjugated actual divergence have leading term $b_0$. Its scalar equation makes the next normal constraint compatible with the vector equation; the final correction will be made only in the physical displacement equation. For $1 \le j \le N$, we seek smooth coefficients $(a_j, b_j)$ on successively smaller disks such that

$$\begin{aligned} R^\theta(a_j, b_j) &= -R(a_{j-1}, b_{j-1}), \\ S^\theta(a_j, b_j) &= -S(a_{j-1}, b_{j-1}), \\ \theta \cdot a_j &= -(\operatorname{div} a_{j-1} - b_{j-1}). \end{aligned} \tag{4.8}$$

Throughout this recursion, write $q_j = \operatorname{div} a_j - b_j$ and set $a_{-1} = b_{-1} = q_{-1} = 0$. The last equation in (4.8) is the displacement–divergence constraint at order $j$.

We first verify the compatibility of these three equations. Conjugating $P$ gives $E_\tau^{-1} P E_\tau = P + \tau P^\theta$. Comparing coefficients of $\tau$ in the conjugated identity (3.2) yields, for arbitrary $(a, b)$,

$$\operatorname{div} R^\theta(a,b) + \theta \cdot R(a,b) - S^\theta(a,b) = P(\theta \cdot a) + P^\theta(\operatorname{div} a - b). \tag{4.9}$$

Suppose the preceding recursive equations have been solved. Apply (4.9) to $(a_{j-1}, b_{j-1})$. For $j \ge 2$, the preceding vector and scalar equations, followed by (3.2), give

$$\begin{aligned} \operatorname{div} R^\theta(a_{j-1}, b_{j-1}) - S^\theta(a_{j-1}, b_{j-1}) &= -\operatorname{div} R(a_{j-2}, b_{j-2}) + S(a_{j-2}, b_{j-2}) \\ &= -P q_{j-2}. \end{aligned}$$

The preceding normal constraint gives the same term on the other side:

$$P(\theta \cdot a_{j-1}) = -P q_{j-2}.$$

For $j = 1$, both expressions are zero by the homogeneous leading equations and $q_{-1} = 0$. In every case these two terms cancel in (4.9), leaving

$$\theta \cdot R(a_{j-1}, b_{j-1}) = P^\theta q_{j-1}. \tag{4.10}$$

Choose a constant vector $n_\theta \in \mathbb{C}^3$ with $\theta \cdot n_\theta = 1$, and set $a_j^{\mathrm{p}} = -q_{j-1} n_\theta$, $b_j^{\mathrm{p}} = 0$. After subtracting this particular pair from the desired $(a_j, b_j)$, the unknown takes values in $\mathcal{E}_\theta$. Equations (4.4) and (4.10) show that the residual vector source belongs to $H_\theta$. Its scalar source is unrestricted. We are therefore left with an inhomogeneous square system on $\mathcal{E}_\theta$ with invertible derivative coefficient. Use the uncut frame $\mathcal{F}$ and the inhomogeneous part of Lemma 4.1 to solve it on $D^2_{R_j} \times I_0$. All source terms, including the particular pair and its derivatives, have transverse support in $K$; the constructed solution retains that support and extends smoothly by zero in $t$. This proves the induction. After finitely many steps, every coefficient is smooth on $D^2_{R_N} \times \mathbb{R}$, hence on a neighborhood of $\overline{B}_{5r}$.

Set

$$a^{[N]} = \sum_{j=0}^{N} \tau^{-j} a_j, \qquad b^{[N]} = \sum_{j=0}^{N} \tau^{-j} b_j.$$

The first equation of (4.8) telescopes to

$$E_\tau^{-1} R(E_\tau a^{[N]}, E_\tau b^{[N]}) = \tau^{-N} R(a_N, b_N). \tag{4.11}$$

The normal constraints telescope separately, giving

$$E_\tau^{-1} \operatorname{div}(E_\tau a^{[N]}) = b^{[N]} + \tau^{-N} q_N. \tag{4.12}$$

Using the dependence of $R$ on its scalar argument, these identities give the physical residual explicitly:

$$\begin{aligned} f_\tau &:= E_\tau^{-1} L(E_\tau a^{[N]}) \\ &= \tau^{-N} \big[ R(a_N, b_N) + k\nabla q_N + (\nabla\lambda) q_N + \tau k\theta q_N \big]. \end{aligned} \tag{4.13}$$

All coefficients on the right are fixed smooth functions. Consequently $\|f_\tau\|_{L^2(B_{5r})} = O(\tau^{1-N})$.

Corollary 3.3 gives a correction $r_\tau \in L^2(B_{5r}; \mathbb{C}^3)$ satisfying

$$E_\tau^{-1} L(E_\tau r_\tau) = -f_\tau, \qquad \|r_\tau\|_{L^2(B_{5r})} = O(\tau^{1-N}).$$

It is smooth in $B_{5r}$ by Lemma 3.4. The same Lemma, on $B_{4r}$, gives

$$\sum_{j=0}^{2} h^j \left\| \nabla^j r_\tau \right\|_{L^2(B_{4r})} = O(\tau^{1-N}). \tag{4.14}$$

Define its conjugated physical divergence by

$$d_\tau = E_\tau^{-1} \operatorname{div}(E_\tau r_\tau) = \operatorname{div} r_\tau + \tau\theta \cdot r_\tau.$$

It follows directly from (4.14) that

$$\|d_\tau\|_{L^2(B_{4r})} + h \|\nabla d_\tau\|_{L^2(B_{4r})} \le C\tau \sum_{j=0}^{2} h^j \left\| \nabla^j r_\tau \right\|_{L^2(B_{4r})} = O(\tau^{2-N}). \tag{4.15}$$

Now put $u_\tau = E_\tau(a^{[N]} + r_\tau)$. It is a smooth exact solution of $Lu_\tau = 0$ on $B_{5r}$, and its physical pair satisfies

$$E_\tau^{-1}(u_\tau, \operatorname{div} u_\tau) = (a^{[N]} + r_\tau, b^{[N]} + \tau^{-N} q_N + d_\tau).$$

The smooth finite expansions and the remainder estimates imply the first assertion of (4.7). They also show that, if the last displayed pair is denoted by $\mathcal{A}_\tau$, then

$$\|\mathcal{A}_\tau\|_{L^2(B_{4r})} + h \|\nabla \mathcal{A}_\tau\|_{L^2(B_{4r})} = O(1).$$

Since, for $j = 1, 2, 3$, $E_\tau^{-1} \partial_j (E_\tau \mathcal{A}_\tau) = \partial_j \mathcal{A}_\tau + \tau\theta_j \mathcal{A}_\tau$, the second assertion of (4.7) follows. Repeating the construction for the three initial columns proves the Proposition. □

## 5 Matching the leading transports

Proposition 5.3 constructs the unique holomorphic comparison of the two leading transports. We first derive a transport normal form and prove a supported estimate for parameter dependence. Throughout this section, a dot between complex vectors denotes the complex bilinear Euclidean product.

**Lemma 5.1** (Transport normal form)**.** *For one coefficient pair, set $g = \nabla \log m$. The change of variables*

$$p = \sqrt{m}\, a, \qquad s = -\frac{k}{2\sqrt{m}}\, b - \frac{1}{2} g \cdot p \tag{5.1}$$

*is a smooth invertible map of $\mathcal{E}_\theta$ to itself. It transforms (4.3) into*

$$D_\theta \begin{pmatrix} p \\ s \end{pmatrix} = M_\theta \begin{pmatrix} p \\ s \end{pmatrix}, \qquad M_\theta = \begin{pmatrix} 0 & \theta \\ Q_\theta \cdot & T_\theta \end{pmatrix} \quad \text{on } H_\theta \oplus \mathbb{C}, \tag{5.2}$$

*where*

$$\begin{aligned} T_\theta &= D_\theta \log(k/\ell), \\ Q_\theta &= \frac{1}{2}\left[ \left( \frac{m}{\ell} D_\theta \log k - \frac{1}{2}\theta \cdot g \right) g - \frac{m}{\ell} D_\theta g \right]. \end{aligned} \tag{5.3}$$

*In particular, the bottom-left entry in (5.2) is the covector $p \mapsto Q_\theta \cdot p$. Both $D_\theta$ and $M_\theta$ depend linearly and holomorphically on $\theta$.*

*Proof.* The inverse of (5.1) is

$$a = m^{-1/2} p, \qquad b = -\frac{2\sqrt{m}}{k} \left( s + \frac{1}{2} g \cdot p \right).$$

It is well defined because $m, k > 0$, and multiplication by $\sqrt{m}$ preserves $H_\theta$. Write $D = D_\theta$. Substitution into the first equation in (4.1) gives

$$0 = 2\sqrt{m}\, Dp + \theta(kb + \sqrt{m}\, g \cdot p), \qquad \text{hence} \qquad Dp = \theta s.$$

Dividing the second equation by two and using $\nabla m = mg$ yields

$$0 = \ell Db + (Dk)b + 2\nabla m \cdot Da + (D\nabla m) \cdot a.$$

Here

$$\begin{aligned} 2\nabla m \cdot Da &= 2\sqrt{m}\, g \cdot Dp - \sqrt{m}\, (D \log m) g \cdot p, \\ (D\nabla m) \cdot a &= \sqrt{m}\, ((D \log m) g + Dg) \cdot p. \end{aligned}$$

The terms containing $(D \log m) g \cdot p$ cancel. Thus

$$\ell Db = -(Dk)b - \sqrt{m}(2(\theta \cdot g)s + (Dg) \cdot p). \tag{5.4}$$

Differentiate the definition of $s$ and substitute (5.4), the expression for $b$, and $Dp = \theta s$. With $c = m/\ell$ and $\gamma = D \log m = \theta \cdot g$, the result is

$$Ds = c(D \log k - \gamma)s + \frac{1}{2}[(cD \log k - \gamma/2)g - cDg] \cdot p.$$

Finally, since $\ell = k + m$,

$$D \log(k/\ell) = D \log k - \frac{Dk + Dm}{\ell} = \frac{m}{\ell}(D \log k - D \log m).$$

This proves the formulas. The top row takes values in $H_\theta$ because $\theta \cdot \theta = 0$. All the direction dependence displayed in the formulas is linear. □

The comparison equation can now be described in terms of the columns that the physical construction must supply. Fix $\theta$ and a cylinder $D^2_{2r} \times J$. Suppose for the moment that $Y_1 : \mathbb{C}^3 \to \mathcal{E}_\theta$ is a smooth invertible matrix of normalized transport columns for the first medium, and that $Y_2 : \mathbb{C}^3 \to \mathcal{E}_\theta$ has locally $L^2$ columns satisfying the second transport equation in distributions:

$$D_\theta Y_j = M_{j,\theta} Y_j, \qquad j = 1, 2.$$

The columns in each $Y_j$ are normalized by that medium's own change of variables (5.1). These changes agree outside $\overline{\Omega}$, where the extended coefficients agree. Thus exterior agreement of the corresponding physical leading columns gives $Y_2 = Y_1$ there.

Under this exterior agreement, define

$$G = Y_2 Y_1^{-1} \in \mathrm{End}(\mathcal{E}_\theta).$$

Only $Y_1$ is required to be invertible. Multiplication by its smooth inverse is legitimate for locally $L^2$ columns, and the product rule in distributions gives

$$D_\theta G = M_{2,\theta} G - G M_{1,\theta}, \qquad G = \mathrm{Id} \quad \text{outside } \overline{\Omega}$$

on the cylinder. Consequently $W = G - \mathrm{Id}$ is locally $L^2$ and supported in $\overline{\Omega}$ within that cylinder.

For a transverse coordinate $t$, restrict the coefficients to the corresponding plane and use a basis of $\mathcal{E}_\theta$ fixed in $x$. On planar endomorphism-valued unknowns define

$$\mathcal{P}_{\theta,t} W = D_\theta W - M_{2,\theta} W + W M_{1,\theta}, \qquad f_{\theta,t} = M_{2,\theta} - M_{1,\theta}. \tag{5.5}$$

The supported planar equation required for the comparison is

$$\mathcal{P}_{\theta,t} W = f_{\theta,t}. \tag{5.6}$$

Matrices here may be regarded as vectors of their nine entries. For almost every $t$, the $L^2$ planar restriction of the local $W$ is supported in $\overline{D^2_r}$ because $\overline{\Omega} \subset B_r$. The physical transfer and weak-limit argument below will supply the second columns $Y_2$. Slicing their distributional cylinder equation will initially give (5.6) only for almost every $t$. We therefore need both uniqueness for supported planar solutions and a way to pass from this almost-everywhere construction to smooth dependence on every parameter. The next lemma provides these facts on fixed Hilbert spaces, which will also allow the planes to vary.

**Lemma 5.2** (Supported planar transport)**.** *Let $\mathcal{U} \subset \mathbb{R}^d$ be open, let $n \geq 1$, and suppose that*

$$\mathcal{P}_\rho = \gamma(\rho)(\partial_{y_1} + i\partial_{y_2}) + A(y, \rho), \qquad \rho \in \mathcal{U},$$

*where $\gamma$ is smooth and nonzero and $A$ is a smooth complex $n \times n$ matrix on a neighborhood of $\overline{D^2_{2r}} \times \mathcal{U}$. Then the following statements hold.*

(i) *A distributional solution of $\mathcal{P}_\rho W = 0$ on $D^2_{2r}$ whose support is a compact subset of that disk is zero.*

(ii) *For every integer $j \geq 0$ and every $W \in H^{j+1}(D^2_{2r}; \mathbb{C}^n)$ supported in $\overline{D^2_r}$,*

$$\|W\|_{H^{j+1}(D^2_{2r})} \leq C_j \, \|\mathcal{P}_\rho W\|_{H^j(D^2_{2r})}, \qquad \operatorname{supp} W \subset \overline{D^2_r}. \tag{5.7}$$

*The constants are locally uniform in $\rho$.*

(iii) *Suppose that $f(y,\rho)$ is smooth on a neighborhood of $\overline{D^2_{2r}}\times\mathcal{U}$. If the equation $\mathcal{P}_\rho W = f(\cdot,\rho)$ has an $L^2$ solution supported in $\overline{D^2_r}$ for a dense set of parameters $\rho$, then it has such a solution for every $\rho$. These solutions are unique and form a smooth function of $(y,\rho)$.*

*The same conclusions apply to matrix-valued unknowns, after their entries are regarded as a vector.*

*Proof.* For a fixed parameter, $\mathcal{P}_\rho$ is elliptic in the two real planar variables: its principal symbol is $\gamma(\rho)(i\xi_1-\xi_2)$, which is nonzero for $\xi\in\mathbb{R}^2\setminus\{0\}$. On each sufficiently small disk, Lemma 4.1 gives a smooth invertible matrix $F$ satisfying $\mathcal{P}_\rho F = 0$. If $\mathcal{P}_\rho W = 0$, then

$$(\partial_{y_1} + i\partial_{y_2})(F^{-1}W) = 0$$

in distributions. Its components are therefore holomorphic functions of $y_1+iy_2$. A homogeneous solution that vanishes on an open set vanishes on every overlapping frame disk by the holomorphic identity theorem. Connectedness propagates this conclusion throughout $D^2_{2r}$. A compactly supported solution vanishes near the boundary of the disk, proving (i).

We give the estimate on the fixed supported space. Extending $W$ by zero to $\mathbb{R}^2$ is harmless because its support lies in $\overline{D^2_r}$. The Fourier symbol just computed gives

$$\|W\|_{H^{j+1}} \le C(\|\gamma(\rho)(\partial_{y_1}+i\partial_{y_2})W\|_{H^j} + \|W\|_{L^2}).$$

Here and below in this proof the norms can equivalently be taken on $D^2_{2r}$, since the functions under consideration are supported in the smaller disk. Smooth multiplication and, for $j\ge 1$, the interpolation estimate $\|W\|_{H^j} \le \varepsilon\|W\|_{H^{j+1}} + C_\varepsilon\|W\|_{L^2}$ give

$$\|W\|_{H^{j+1}} \le C(\|\mathcal{P}_\rho W\|_{H^j} + \|W\|_{L^2}). \tag{5.8}$$

The coefficients can be cut off outside a neighborhood of the smaller disk for this calculation. We will also use local regularity for an $L^2$ distributional solution with smooth right side. In a local frame $F$, its equation becomes a scalar Cauchy–Riemann equation componentwise with smooth right side. Subtracting a smooth local particular solution, as supplied by Lemma 4.1, leaves holomorphic components. The original solution is therefore smooth in the disk.

To remove the last term in (5.8), suppose that (5.7) fails at a fixed parameter. There would then be a sequence $W_\nu$, supported in $\overline{D^2_r}$, such that

$$\|W_\nu\|_{H^{j+1}} = 1, \qquad \|\mathcal{P}_\rho W_\nu\|_{H^j} \longrightarrow 0.$$

After passing to a subsequence, compactness of the embedding into $L^2(D^2_{2r})$ gives a strong $L^2$ limit $W$. This limit is supported in $\overline{D^2_r}$ and solves $\mathcal{P}_\rho W = 0$ distributionally. Part (i) gives $W = 0$. Equation (5.8) now contradicts the normalization of $W_\nu$. This proves (ii) at a fixed parameter.

For the parameter assertions, equip the fixed complex Hilbert spaces

$$X_j = \{W\in H^{j+1}(D^2_{2r};\mathbb{C}^n) : \operatorname{supp} W \subset \overline{D^2_r}\},$$
$$Z_j = H^j(D^2_{2r};\mathbb{C}^n)$$

with their usual Hermitian Sobolev inner products. The subspace $X_j$ is closed. The map $\rho\mapsto\mathcal{P}_\rho$ is smooth in $\mathcal{L}(X_j,Z_j)$. The estimate at $\rho_0$ persists in a neighborhood: absorb $\|(\mathcal{P}_\rho-\mathcal{P}_{\rho_0})W\|_{Z_j}$ into its left side when the operator norm of the difference is small. This proves local uniformity in (ii).

Write $\mathcal{P}_\rho^* : Z_j\to X_j$ for the Hilbert space adjoint with respect to these fixed inner products. In particular, this is not an adjoint in the complex bilinear pairing used for the null vectors. The lower bound in (ii) implies that $\mathcal{P}_\rho^*\mathcal{P}_\rho : X_j\to X_j$ is coercive and invertible. Consequently

$$T_{\rho,j} = (\mathcal{P}_\rho^*\mathcal{P}_\rho)^{-1}\mathcal{P}_\rho^* : Z_j \longrightarrow X_j \tag{5.9}$$

is a bounded left inverse. It depends smoothly on $\rho$, since adjoints and inversion of bounded invertible operators do so locally. This argument uses a lower bound for $\mathcal{P}_\rho$; it requires no surjectivity onto $Z_j$.

For each $j$, set $W_j(\rho) = T_{\rho,j} f(\cdot, \rho)$. An $L^2$ supported solution, whenever it exists, is smooth by interior ellipticity and hence belongs to every $X_j$. It must equal $W_j(\rho)$ by the left-inverse identity. Thus the smooth $Z_j$-valued residual

$$\mathcal{P}_\rho W_j(\rho) - f(\cdot, \rho)$$

vanishes on the assumed dense set, and continuity makes it zero for every $\rho$. Uniqueness in (i) identifies the solutions for different $j$. Smooth dependence with values in every $H^{j+1}$, together with Sobolev embedding in the planar variables, gives joint smoothness in $(y, \rho)$. This proves (iii). □

Let

$$\mathcal{C} = \{[\theta] \in \mathbb{C}\mathrm{P}^2 : \theta \cdot \theta = 0\}$$

be the projective null conic. The fibers $\mathcal{E}_\theta = H_\theta \oplus \mathbb{C}$ form a holomorphic subbundle $\mathcal{E}$ of the trivial bundle $\mathcal{C} \times \mathbb{C}^4$. For example, on a chart where $\theta_i \neq 0$, the vectors $e_j - (\theta_j/\theta_i)e_i$, $j \neq i$, give a holomorphic frame of $H_\theta$; adjoining the scalar component gives one of $\mathcal{E}$.

**Proposition 5.3** (Matching of the transports)**.** *For the two extended coefficient pairs of Lemma 2.1, equality of the physical displacement-to-traction maps implies that there is a unique smooth field*

$$G(x, [\theta]) \in \mathrm{End}(\mathcal{E}_\theta), \qquad (x, [\theta]) \in \mathbb{R}^3 \times \mathcal{C},$$

*satisfying, for every nonzero representative $\theta$,*

$$\begin{aligned} D_\theta G &= M_{2,\theta} G - G M_{1,\theta}, \\ G(x, [\theta]) &= \mathrm{Id} \qquad \text{if } |x| > 2r. \end{aligned} \tag{5.10}$$

*For each fixed $x$, this field is a holomorphic section of* $\mathrm{End}(\mathcal{E})$ *over $\mathcal{C}$. In fact, for each fixed direction the first equation has at most one smooth solution whose difference from the identity has compact spatial support.*

*Proof. Physical transfer and weak limits.* Fix a nonzero null vector $\theta$ and $t_0 \in (-2r, 2r)$. Apply Proposition 4.2 to the first coefficient pair. It supplies three exact smooth solutions on $B_{5r}$, with the limits and estimates in (4.7), and an interval $J$ containing $t_0$ on which their leading columns form a frame on $D^2_{2r} \times J$. We perform the following construction for each column, temporarily suppressing its index.

By Proposition 2.2, the first solution $u_{1,\tau}$ has a smooth transferred solution $u_{2,\tau}$ on $B_{5r}$, with $u_{2,\tau} = u_{1,\tau}$ outside $\overline{\Omega}$. Define the physical augmented vectors

$$V_{j,\tau} = (u_{j,\tau}, \operatorname{div} u_{j,\tau}), \qquad j = 1, 2.$$

The difference $V_{2,\tau} - V_{1,\tau}$ is smooth and compactly supported in $\overline{\Omega} \subset B_r$. By Lemma 3.1,

$$\mathcal{L}_2(V_{2,\tau} - V_{1,\tau}) = (\mathcal{L}_1 - \mathcal{L}_2) V_{1,\tau}. \tag{5.11}$$

Both normalized systems have principal part $\Delta \mathrm{Id}_4$. Their difference is therefore an operator of order at most one, whose coefficients are supported in $\overline{\Omega}$. The weighted derivative bound in (4.7) implies

$$\begin{aligned} \left\| E_\tau^{-1} (\mathcal{L}_1 - \mathcal{L}_2) V_{1,\tau} \right\|_{L^2} &\le C \left( \left\| E_\tau^{-1} V_{1,\tau} \right\|_{L^2(B_{4r})} + \left\| E_\tau^{-1} \nabla V_{1,\tau} \right\|_{L^2(B_{4r})} \right) \\ &= O(\tau). \end{aligned}$$

The function on the left is extended by zero off its compact support when a norm on $\mathbb{R}^3$ is used. Apply Proposition 3.2 to the difference in (5.11). Since $\|F\|_{H_h^{-1}} \le \|F\|_{L^2}$ and $h = \tau^{-1}$, we obtain

$$\begin{aligned} \left\| E_\tau^{-1}(V_{2,\tau} - V_{1,\tau}) \right\|_{H_h^1} &\le Ch \left\| E_\tau^{-1}(\mathcal{L}_1 - \mathcal{L}_2) V_{1,\tau} \right\|_{H_h^{-1}} \\ &\le C. \end{aligned} \tag{5.12}$$

Thus $E_\tau^{-1} V_{2,\tau}$ is bounded in $L^2(B_{4r})$. After passage to a subsequence, chosen simultaneously for the three columns, it converges weakly to a vector $(\widetilde{a}, \widetilde{b})$. The strong convergence of the first amplitudes and the exterior agreement of the physical solutions give

$$(\widetilde{a}, \widetilde{b}) = (a_0, b_0) \quad \text{almost everywhere on } B_{4r} \setminus \overline{\Omega}. \tag{5.13}$$

These weak limits satisfy the constrained transport for the second pair. To see this directly, write $(a_\tau, b_\tau) = E_\tau^{-1} V_{2,\tau}$. The exact equations and the null condition give

$$\begin{aligned} R_2^\theta(a_\tau, b_\tau) + \tau^{-1} R_2(a_\tau, b_\tau) &= 0, \\ S_2^\theta(a_\tau, b_\tau) + \tau^{-1} S_2(a_\tau, b_\tau) &= 0. \end{aligned}$$

When tested against a fixed compactly supported smooth function, the terms multiplied by $\tau^{-1}$ tend to zero: all their derivatives can be transferred to that test function, while the amplitudes remain bounded in $L^2$. Weak convergence passes the remaining terms to the limit. Moreover, the physical divergence identity says

$$\theta \cdot a_\tau = \tau^{-1}(b_\tau - \operatorname{div} a_\tau),$$

so $\theta \cdot \widetilde{a} = 0$ distributionally. We have proved

$$R_2^\theta(\widetilde{a}, \widetilde{b}) = 0, \qquad S_2^\theta(\widetilde{a}, \widetilde{b}) = 0, \qquad (\widetilde{a}, \widetilde{b}) \in \mathcal{E}_\theta.$$

No derivative estimate on the transferred amplitudes, beyond their $L^2$ bound, is needed for this passage to the limit.

The physical transfer has therefore produced the second transport columns, but only as weak limits. We next obtain a supported comparison on almost every transverse plane and use supported uniqueness to extend it to every plane.

*A comparison on almost every plane.* Apply (5.1) to the three leading columns for each pair and denote the resulting maps by $Y_j : \mathbb{C}^3 \to \mathcal{E}_\theta$. For $j = 2$, these columns are initially locally $L^2$; they satisfy the equations distributionally. Since $J \subset (-2r, 2r)$, the cylinder on which the first columns form a frame satisfies

$$D_{2r}^2 \times J \subset B_{\sqrt{8}\,r} \subset B_{4r}.$$

Thus the weak limits constructed above are defined throughout this cylinder, where

$$D_\theta Y_j = M_{j,\theta} Y_j,$$

and $Y_1$ is smooth and invertible. The change of variables agrees for the two pairs outside $\overline{\Omega}$. Define

$$G = Y_2 Y_1^{-1}.$$

The calculation preceding Lemma 5.2 now applies: $G$ is a locally $L^2$ endomorphism on the cylinder, equals the identity outside $\overline{\Omega}$, and satisfies the first equation of (5.10) distributionally. With

$W = G - \mathrm{Id}$, this is the cylinder version of (5.6). The operator $D_\theta$ differentiates only the planar variables. Testing the cylinder equation with products of a planar test function and a transverse test function, and then using a countable dense collection of planar tests, gives (5.6) on $D^2_{2r}$ for almost every $t \in J$. For these $t$, $W(\cdot, t)$ belongs to $L^2(D^2_{2r})$ and has support in $\overline{D^2_r}$. Indeed, $W = 0$ when $|y| > r$, and on its remaining support multiplication by $Y_1^{-1}$ is locally bounded uniformly in $t$. These assertions can first be made on compact subintervals of $J$ and then exhausted over $J$.

*Existence on every plane.* In the orthonormal coordinates associated with the fixed $\theta$,

$$D_\theta = |\operatorname{Re}\theta|(\partial_{y_1} + i\partial_{y_2}).$$

Thus (5.5), as an operator on matrix entries, has the form of Lemma 5.2. Its coefficients and the source $f_{\theta,t}$ are smooth in $(y, t)$. The set of parameters for which a supported solution has just been obtained has full measure in $J$, and hence is dense. Part (iii) of that lemma extends existence to every $t \in J$ and makes $W$ smooth in $(y, t)$. It also makes the choice unique, so the results from different intervals $J$ agree on their overlaps.

Since $t_0$ was arbitrary, this constructs $G$ for every $-2r < t < 2r$. Extend $W = G - \mathrm{Id}$ by zero in the planar variables outside $D^2_{2r}$. This extension is smooth because $W$ is already zero for $r < |y| < 2r$. The equation holds across this extension, and it holds farther out because $f_{\theta,t} = 0$ there. If $|t| > r$, the entire plane misses $\overline{\Omega}$, so $f_{\theta,t} = 0$. Part (i) of Lemma 5.2 then gives $W = 0$ on that plane. Extending by zero also for $|t| \geq 2r$ is therefore smooth. For this fixed direction we have constructed a global smooth comparison with

$$\operatorname{supp}(G(\cdot, \theta) - \mathrm{Id}) \subset \{(y, t) : |y| \leq r,\ |t| \leq r\} \subset \overline{B_{2r}}. \tag{5.14}$$

The final containing ball is independent of the direction.

For later use, uniqueness holds even if a larger compact support is allowed. The difference of two such comparisons solves the homogeneous equation on every plane and has compact support there. The local-frame and holomorphic-continuation argument in Lemma 5.2, applied on a disk containing that support, makes the difference zero on each plane.

*Smooth dependence on the direction.* The preceding construction gives a unique comparison separately for every nonzero null vector $\theta$. If $c \in \mathbb{C} \setminus \{0\}$, then $\mathcal{E}_{c\theta} = \mathcal{E}_\theta$, and

$$D_{c\theta} = cD_\theta, \qquad M_{j,c\theta} = cM_{j,\theta}.$$

The equations for $\theta$ and $c\theta$ are consequently identical after division by $c$. Compact-support uniqueness gives $G(x, c\theta) = G(x, \theta)$, so the field is already well defined on projective directions as a pointwise family.

Choose a local smooth representative $\theta(v)$ of those directions, where $v$ ranges over an open subset of $\mathbb{R}^2$, and a smooth local frame of $\mathcal{E}$. Define the orthonormal vectors

$$e_1(v) = \frac{\operatorname{Re}\theta(v)}{|\operatorname{Re}\theta(v)|}, \quad e_2(v) = \frac{\operatorname{Im}\theta(v)}{|\operatorname{Im}\theta(v)|}, \quad e_3(v) = e_1(v) \times e_2(v).$$

They vary smoothly: the real and imaginary parts of a nonzero null vector are nonzero, perpendicular, and of equal length. In the coordinates

$$x = y_1 e_1(v) + y_2 e_2(v) + t e_3(v),$$

and the chosen bundle frame, (5.6) is a smooth family of operators of the form in Lemma 5.2, with parameters $(t, v)$ on the fixed disk $D^2_{2r}$. For every value of these parameters a solution supported

in $\overline{D_r^2}$ has already been constructed. The smooth left inverse (5.9) therefore expresses this unique solution as a smooth function of $(y,t,v)$. Returning to physical coordinates gives joint smoothness in $(x,v)$.

This reasoning uses the separate fixed-direction existence results and their uniqueness. In particular, none of the earlier weakly convergent subsequences has to be chosen simultaneously for varying directions.

*Holomorphic dependence at fixed physical coordinates.* Now let $v$ be a complex local coordinate on $\mathcal{C}$. Choose a holomorphic nonzero representative $\theta(v)$ and a holomorphic frame of $\mathcal{E}$ on this chart. This frame depends only on $v$. In this frame the matrices $M_{j,\theta(v)}(x)$ are holomorphic in $v$ for each fixed $x$: the ambient formulas (5.2)–(5.3) are holomorphic in $\theta$, and passage to a holomorphic bundle frame preserves that property.

Represent $G$ in the same frame and differentiate the first equation of (5.10) by $\partial_{\bar v}$, *holding* $x\in\mathbb{R}^3$ *fixed*. Joint smoothness just established justifies this differentiation. Since the representative and the frame are holomorphic and the frame is independent of $x$, the result is exactly

$$D_{\theta(v)}(\partial_{\bar v}G) - M_{2,\theta(v)}\partial_{\bar v}G + (\partial_{\bar v}G)M_{1,\theta(v)} = 0.$$

The derivative here is a matrix in a holomorphic trivialization, hence represents an endomorphism of the same fiber $\mathcal{E}_{\theta(v)}$. The common support bound (5.14) implies that this derivative is zero for $|x|>2r$. Compact-support uniqueness on each plane gives $\partial_{\bar v}G=0$. The smooth moving coordinates were used to establish joint regularity; the equation was differentiated only after returning to fixed physical coordinates. We conclude that $G(x,\cdot)$ is holomorphic on $\mathcal{C}$ for every $x$, completing the proof. □

# 6 Recovery of the coefficients

The holomorphic comparison from Proposition 5.3 is constrained by the geometry of its polarization bundle. We first show that this comparison is the identity, and then recover the coefficients from the transport matrices.

## 6.1 The bundle on the null conic

Let $H\to\mathcal{C}$ be the holomorphic bundle with fiber $H_\theta$, and let $\mathcal{N}\subset H$ be the null line subbundle with fiber $\mathbb{C}\theta$. Thus $\mathcal{E}=H\oplus\mathcal{O}$, where $\mathcal{O}$ denotes the trivial holomorphic line bundle on $\mathcal{C}$. On $\mathbb{P}^1(\mathbb{C})$, let $\mathcal{O}(-1)$ denote the tautological line bundle and $\mathcal{O}(-2)$ its tensor square.

**Lemma 6.1.** *The bundle $H$ has no nonzero global holomorphic sections, and $H/\mathcal{N}$ is holomorphically trivial. More precisely, under the identification $\mathcal{C}\simeq\mathbb{P}^1(\mathbb{C})$,*

$$H\simeq\mathcal{O}(-1)^{\oplus 2},\qquad \mathcal{N}\simeq\mathcal{O}(-2),$$

*and their quotient sequence is*

$$0\longrightarrow\mathcal{O}(-2)\xrightarrow{(z_0,z_1)^T}\mathcal{O}(-1)^{\oplus 2}\xrightarrow{(-z_1,z_0)}\mathcal{O}\longrightarrow 0. \tag{6.1}$$

*Here $[z_0:z_1]$ are homogeneous coordinates on $\mathbb{P}^1(\mathbb{C})$.*

*Proof.* A homogeneous parametrization of the conic is

$$\theta(z_0,z_1)=(z_0^2-z_1^2,\ i(z_0^2+z_1^2),\ 2z_0z_1). \tag{6.2}$$

It identifies $\mathbb{P}^1(\mathbb{C})$ with $\mathcal{C}$: on the first coordinate chart its inverse is $z_1/z_0 = \theta_3/(\theta_1 - i\theta_2)$, and on the second it is $z_0/z_1 = \theta_3/(-\theta_1 - i\theta_2)$. The denominators cannot both vanish at a point of $\mathcal{C}$.

Consider the two degree-one columns

$$h_1 = (z_0, iz_0, z_1), \qquad h_2 = (-z_1, iz_1, z_0).$$

Direct calculation, with the complex bilinear product, gives

$$\theta \cdot h_1 = \theta \cdot h_2 = 0, \qquad h_1 \times h_2 = i\theta, \qquad \theta = z_0 h_1 + z_1 h_2. \tag{6.3}$$

To interpret these homogeneous formulas as bundle maps, write $\mathcal{O}(-1)_{[z]} = \mathbb{C}z$ for $z = (z_0, z_1)$. The assignment

$$(\xi_1 z, \xi_2 z) \longmapsto \xi_1 h_1(z) + \xi_2 h_2(z)$$

is independent of the representative: replacing $z$ by $cz$ replaces $\xi_j$ by $c^{-1}\xi_j$ and $h_j(z)$ by $ch_j(z)$. Its local polynomial formulas are holomorphic. Since the images lie in $H$ and are linearly independent at every point, it is an isomorphism $\mathcal{O}(-1)^{\oplus 2} \to H$. Likewise, $\xi z^{\otimes 2} \mapsto \xi\theta(z)$ is well defined because $\theta(cz) = c^2\theta(z)$, and identifies $\mathcal{O}(-2)$ with $\mathcal{N}$. The final identity in (6.3) gives the first map in (6.1).

The expression $-z_1\xi_1 + z_0\xi_2$ is also unchanged under the same representative scaling, so its local polynomial formulas define a holomorphic map to the trivial line bundle $\mathcal{O}$. The row $(-z_1, z_0)$ is everywhere surjective, and its kernel is the image of $(z_0, z_1)^T$. This proves the quotient assertion, including at both coordinate-chart endpoints. It is also consistent with the induced bilinear form: for $p = \xi_1 h_1 + \xi_2 h_2$ in a local frame,

$$p \cdot p = (-z_1\xi_1 + z_0\xi_2)^2.$$

Finally, any holomorphic section of $H$ has holomorphic ambient components on the compact conic, so those components are constant. Such a constant vector is orthogonal to every null vector. The vectors $\theta(1,0)$, $\theta(0,1)$ and $\theta(1,1)$ span $\mathbb{C}^3$, so the section is zero. □

*Remark* 6.2. The exact sequence (6.1) does not split holomorphically: a splitting would give a nonzero global section of $H$. On the other hand, the displayed splitting of $H$ itself gives nonscalar holomorphic endomorphisms of $H$. The next argument uses the transport equation to restrict those endomorphisms.

**Proposition 6.3.** *The comparison endomorphism of Proposition 5.3 is* $G = \mathrm{Id}$. *Consequently, for every* $x \in \mathbb{R}^3$ *and every nonzero null vector* $\theta$,

$$M_{1,\theta} = M_{2,\theta} \quad \text{on } \mathcal{E}_\theta. \tag{6.4}$$

*Proof.* Fix $x$. Relative to $\mathcal{E} = H \oplus \mathcal{O}$, the upper-right block of the holomorphic endomorphism $G(x, \cdot)$ is a global holomorphic section of $H$, and therefore vanishes by Lemma 6.1. Its lower-right block is a holomorphic scalar function on the compact conic, and hence is independent of direction. We may thus write

$$G = \begin{pmatrix} F & 0 \\ A & f \end{pmatrix}, \qquad f = f(x).$$

All these blocks are smooth in $x$. Taking the upper-right block of (5.10), and using (5.2), gives

$$0 = \theta f - F\theta.$$

It follows that $F - f\mathrm{Id}_H$ vanishes on $\mathcal{N}$. This holomorphic bundle morphism consequently factors through $H/\mathcal{N}$. By Lemma 6.1, its factor is a morphism $\mathcal{O} \to H$, which must vanish. Thus $F = f\mathrm{Id}_H$.

The upper-left block of (5.10) now reads

$$(D_\theta f)\mathrm{Id}_{H_\theta} = \theta A. \tag{6.5}$$

The right side has rank at most one, while $H_\theta$ has dimension two. Therefore $D_\theta f = 0$. Equation (6.5) then gives $A = 0$, since $\theta \neq 0$. As null vectors span $\mathbb{C}^3$, we obtain $\nabla f = 0$. The exterior identity value of $G$ fixes $f = 1$, so $G = \mathrm{Id}$. Substitution in (5.10) proves (6.4). □

### 6.2 A finite-type uniqueness lemma

We will show that equality of the restricted bottom-left transport blocks leaves at most a scalar-identity ambiguity, then use the following elementary lemma to remove it. Indices in the lemma range from 1 to $n$; all sums are written explicitly.

**Lemma 6.4.** *Let $U \subset \mathbb{R}^n$ be connected and open, with $n \geq 2$, and let $\eta, q$ and $B_{ij}{}^a$ be smooth, real- or complex-valued functions on $U$ satisfying*

$$\partial_i \partial_j \eta = \sum_{a=1}^{n} B_{ij}{}^a \partial_a \eta + \delta_{ij} q. \tag{6.6}$$

*Then $(\nabla\eta, q)$ satisfies a homogeneous first-order system with smooth coefficients. In particular, if $\eta$ vanishes on a nonempty open subset of $U$, then $\eta = q = 0$ on $U$.*

*Proof.* Put $v_a = \partial_a \eta$. For each $s$, choose one index $i = i(s)$ with $i \neq s$, without summing over $i$. Equation (6.6) and commutation of third derivatives give

$$\begin{aligned} \partial_s q &= \partial_i \left( \sum_{a=1}^{n} B_{si}{}^a v_a \right) - \partial_s \left( \sum_{a=1}^{n} B_{ii}{}^a v_a \right) \\ &= \sum_{a=1}^{n} (\partial_i B_{si}{}^a - \partial_s B_{ii}{}^a) v_a + \sum_{a=1}^{n} (B_{si}{}^a \partial_i v_a - B_{ii}{}^a \partial_s v_a). \end{aligned}$$

There is no derivative of $q$ on the right because the off-diagonal entry $\partial_s \partial_i \eta$ in (6.6) has $\delta_{si} = 0$. Substituting that equation for the derivatives of $v$ yields

$$\begin{aligned} \partial_s v_a &= \sum_{b=1}^{n} B_{as}{}^b v_b + \delta_{as} q, \\ \partial_s q &= \sum_{b=1}^{n} C_s{}^b v_b + d_s q, \end{aligned} \tag{6.7}$$

where the smooth coefficients are explicitly

$$\begin{aligned} C_s{}^b &= \partial_i B_{si}{}^b - \partial_s B_{ii}{}^b + \sum_{a=1}^{n} (B_{si}{}^a B_{ai}{}^b - B_{ii}{}^a B_{as}{}^b), \\ d_s &= B_{si}{}^i - B_{ii}{}^s. \end{aligned} \tag{6.8}$$

This is the claimed homogeneous system for $v$ and $q$.

If $\eta = 0$ on an open subset, then $v = 0$ there, and a diagonal entry of (6.6) also gives $q = 0$ there. Along any piecewise smooth path in $U$, Equation (6.7) restricts to a homogeneous linear ordinary differential equation for $(v, q)$. Uniqueness with zero initial data propagates its vanishing along the path. An open connected subset of $\mathbb{R}^n$ is path connected, so $v = q = 0$ on $U$. Finally, $\eta$ is constant on $U$, and its value on the initial open subset is zero. □

### 6.3 Completion of the coefficient recovery

*Proof of Theorem 1.1.* Suppose that the two displacement-to-traction maps agree, and use the common exterior extensions furnished by Lemma 2.1. Propositions 5.3 and 6.3 give equality of the transport matrices on $\mathcal{E}_\theta$ for every nonzero null $\theta$. In particular, (5.3) gives

$$D_\theta \log(k_1/\ell_1) = D_\theta \log(k_2/\ell_2).$$

Since null vectors span $\mathbb{C}^3$, the difference of these logarithms has zero gradient. Its exterior value is zero. Hence $k_1/\ell_1 = k_2/\ell_2$ on $\mathbb{R}^3$, and the functions

$$c = \frac{m_1}{\ell_1} = \frac{m_2}{\ell_2} > 0, \qquad b_* = \log \frac{k_1}{m_1} = \log \frac{k_2}{m_2} \tag{6.9}$$

are well defined and smooth. Indeed, $m_j/\ell_j = 1 - k_j/\ell_j$, and $k_j/m_j = (k_j/\ell_j)/(1 - k_j/\ell_j)$.

Equality of the bottom-left blocks on $\mathcal{E}_\theta$ means

$$(Q_{1,\theta} - Q_{2,\theta}) \cdot p = 0 \qquad (p \in H_\theta).$$

The annihilator of $H_\theta = \theta^\perp$ for the nondegenerate ambient bilinear form is $\mathbb{C}\theta$. Since each $Q_{j,\theta}$ is linear in $\theta$, there is a smooth ambient matrix $K(x)$ such that

$$Q_{1,\theta} - Q_{2,\theta} = K(x)\theta \in \mathbb{C}\theta \qquad (\theta \cdot \theta = 0). \tag{6.10}$$

For fixed $x$, the proportionality factor in (6.10) is a holomorphic function on $\mathcal{C}$. To see this at every point, choose a local holomorphic representative $\theta$ and a component $\theta_a \neq 0$; the factor is $(K\theta)_a/\theta_a$. These expressions agree on overlapping charts and are unchanged by rescaling the representative. Compactness of $\mathcal{C}$ makes the factor constant in direction. The spanning property of null vectors then gives

$$K(x) = \kappa(x)\mathrm{Id}, \qquad \kappa(x) = \tfrac{1}{3} \operatorname{tr} K(x). \tag{6.11}$$

In particular, $\kappa$ is smooth.

Set

$$\eta = \log m_1 - \log m_2, \qquad v = \nabla\eta = g_1 - g_2, \qquad \beta = \nabla b_*, \qquad \alpha = \frac{c - 1/2}{c},$$

where $g_j = \nabla \log m_j$. Because $\nabla \log k_j = g_j + \beta$, the expression for $Q$ in (5.3) becomes

$$2Q_{j,\theta} = ((c - 1/2)g_j g_j^T + c g_j \beta^T - c\nabla^2 \log m_j)\theta.$$

Subtract these identities, use $g_1 g_1^T - g_2 g_2^T = g_1 v^T + v g_2^T$, and apply (6.11). Defining the smooth scalar $q = -2\kappa/c$, we obtain

$$\nabla^2\eta = \alpha(g_1 v^T + v g_2^T) + v\beta^T + q\mathrm{Id}.$$

Equivalently,

$$\begin{aligned} \partial_i\partial_j\eta &= \sum_{a=1}^{3} B_{ij}^{\ a}\partial_a\eta + \delta_{ij}q, \\ B_{ij}^{\ a} &= \alpha(g_1)_i\delta_{ja} + (\alpha(g_2)_j + \beta_j)\delta_{ia}. \end{aligned} \tag{6.12}$$

The coefficients $B_{ij}^{\ a}$ are smooth on $\mathbb{R}^3$. They are fixed functions determined by the two coefficient pairs under comparison. The common exterior extension gives $\eta = 0$ on a nonempty exterior open set. Lemma 6.4, applied to (6.12), yields $\eta = 0$ throughout $\mathbb{R}^3$. Thus $m_1 = m_2$. Equation (6.9) then gives $k_1 = m_1 e^{b_*} = m_2 e^{b_*} = k_2$, and therefore

$$\mu_1 = \mu_2, \qquad \lambda_1 = k_1 - m_1 = k_2 - m_2 = \lambda_2.$$

Restricting to $\Omega$ proves the theorem. □

# A Analytic preliminaries and local elliptic estimates

We record the functional, boundary, planar and elliptic facts used in the proof. The integral arguments use Fubini for integrable products, dominated convergence with an integrable majorant, and monotone convergence and Tonelli for nonnegative functions; see the source statements in [6]. Integration by parts follows from the fundamental theorem of calculus and Fubini, first in rectangles and then in smooth local boundary charts. The Fourier inputs are the four explicitly identified results of Salo. Dividing his transform by $(2\pi)^{d/2}$ gives our unitary transform; multiplication by that nonzero constant preserves invertibility and distributional continuity. His derivative rule, with $D_j = -i\partial_j$, becomes $\widehat{\partial_j u} = i\xi_j \widehat{u}$. His modulation rule gives $\widehat{e^{ib\cdot x}u}(\xi) = \widehat{u}(\xi - b)$. These normalization changes are the only changes to those imported Fourier statements.

## A.1 Hilbert, Sobolev and boundary facts

**Lemma A.1** (Hilbert bridges)**.** *A bounded linear functional on a separable complex Hilbert space, with inner product linear in its first variable, has a unique representing vector and the same norm. A bounded coercive sesquilinear form has a unique solution for each bounded antilinear right side. Bounded sequences in separable Hilbert spaces have weakly convergent subsequences. These conclusions apply to the $L^2$ and Sobolev spaces used here.*

*Proof.* Choose a countable orthonormal basis by Gram–Schmidt from a countable dense set, omitting zero residuals. If $\ell$ is bounded, its values on finite orthonormal sums show that $\sum_j |\ell(e_j)|^2 \le \|\ell\|^2$. The convergent series $z = \sum_j \overline{\ell(e_j)} e_j$ therefore gives $\ell(v) = (v, z)$, first on finite sums and then by density. Equality of norms follows by evaluating at $z/\|z\|$ when $z \neq 0$. A closed subspace is Hilbert. Its orthogonal projection exists: a distance-minimizing sequence is Cauchy by the parallelogram identity, and its limit is orthogonal to the subspace by varying in any of its directions.

Represent a bounded sesquilinear form by a bounded operator $T$. If $\operatorname{Re}(Tu, u) \ge c\|u\|^2$, then $\|Tu\| \ge c\|u\|$; thus its range is closed. The same inequality for $T^*$ gives $\ker T^* = 0$, so the orthogonal complement of its range is zero. The range is consequently the whole space, and the claimed solution and bound follow. This proves the coercive-form assertion without an additional solvability theorem. For a bounded sequence, choose a diagonal subsequence on which every orthonormal coefficient converges. The limiting coefficients are square summable with the same bound, by the finite-sum inequalities. Finite-sum approximation and Cauchy–Schwarz then give weak convergence.

For completeness of $L^2$, from any Cauchy sequence select a subsequence whose successive differences have norms at most $2^{-j}$. Finite sums of their absolute values have uniformly bounded $L^2$ norm by the triangle inequality. Their increasing pointwise limit has the same bound by monotone convergence of their squares. Thus the difference series converges absolutely almost everywhere and its tail has $L^2$ norm at most $\sum_{j \ge N} 2^{-j}$; the original sequence has this same $L^2$ limit. Simple functions on rational boxes give separability, using approximation in measure and truncation; alternatively use the Schwartz density in Salo's Lemma 3.4.6. Restriction to a measurable set is a closed subspace after zero extension. Weighted Fourier spaces are Hilbert by Plancherel, and the derivative Sobolev spaces are closed graph subspaces of finite products of $L^2$. Their separability follows from these embeddings. □

**Lemma A.2** (Sobolev bridges)**.** *Multiplication by a fixed smooth compactly supported function is bounded on $H^s(\mathbb{R}^d)$ for every real $s$. $H^s(\mathbb{R}^d)$ embeds into $C^k$ if $s > k + d/2$. For a fixed compact support, the inclusion of a bounded subset of $H^1$ into $L^2$ is compact. For integers $j \ge 1$ and $\epsilon > 0$, $\|v\|_{H^j} \le \epsilon \|v\|_{H^{j+1}} + C_\epsilon \|v\|_2$. These assertions hold componentwise and locally after cutoffs.*

*Proof.* For a smooth compact multiplier $a$, its Fourier transform decreases faster than any power, by integration by parts. The inequality $\langle\eta\rangle^s\langle\xi\rangle^{-s} \le C_s\langle\eta-\xi\rangle^{|s|}$ makes the Fourier kernel of multiplication integrably dominated by a convolution kernel. Cauchy–Schwarz with that kernel and Fubini give the asserted bound. For embedding, $\xi^\alpha\widehat{v}\in L^1$ when $|\alpha|\le k$, by Cauchy–Schwarz and $\int\langle\xi\rangle^{-2(s-k)}\,\mathrm{d}\xi<\infty$. The inverse Fourier integral and dominated convergence then give continuous bounded derivatives through order $k$. Interpolation follows by splitting at a frequency $R$ chosen so that $\langle\xi\rangle^j\le\epsilon\langle\xi\rangle^{j+1}$ for $|\xi|>R$.

For compactness, high-frequency tails of an $H^1$-bounded set have $L^2$ norm at most $C/R$. On a fixed compact physical support $K$ and the frequency ball $|\xi|\le R$, the Fourier kernel $(2\pi)^{-d/2}e^{-ix\cdot\xi}$ is square integrable on the product. Approximation of this kernel in $L^2(K\times B_R)$ by finite sums of rectangle indicators gives finite-rank operators; Cauchy–Schwarz bounds the operator error by the kernel error. The truncated transform is thus compact. A diagonal subsequence for increasing $R$, with the uniform tail bound, is Cauchy in full $L^2$. Plancherel completes the proof. □

**Lemma A.3** (Trace, extension and Dirichlet solution)**.** *On a bounded smooth domain the trace maps $H^1$ onto $H^{1/2}$ with a bounded right inverse, its kernel is $H^1_0$, and the static elastic Dirichlet problem with the positivity in the theorem has a unique weak solution. A function smooth up to this boundary has a smooth neighborhood extension. Weak gluing across the boundary of two $H^1$ fields with the same trace is legitimate.*

*Proof.* In the upper half-space let $\rho=\langle\xi\rangle$, where $\xi$ is tangential frequency. Integration of $-\partial_t|\widehat{u}(\xi,t)|^2$ and $2ab\le a^2+b^2$ give

$$\int\rho|\widehat{u}(\xi,0)|^2\,\mathrm{d}\xi\le\int_0^\infty\int(\rho^2|\widehat{u}|^2+|\partial_t\widehat{u}|^2)\,\mathrm{d}\xi\,\mathrm{d}t.$$

Conversely $\widehat{u}(\xi,t)=e^{-\rho t}\widehat{f}(\xi)$ has the right trace and $H^1$ norm squared equal to $\int\rho|\widehat{f}|^2\,\mathrm{d}\xi$. Localization in $t$ preserves the bound. Flatten finitely many smooth boundary charts and use a subordinate smooth partition of unity to transfer these bounds. The boundary $H^{1/2}$ norm is invariant up to constants under these chart changes: for smooth compactly supported $f$, Fourier and Fubini identify its homogeneous squared seminorm, up to a positive dimension constant, with

$$\iint\frac{|f(x)-f(y)|^2}{|x-y|^{d+1}}\,\mathrm{d}x\,\mathrm{d}y.$$

The identity follows by setting $y=x+s$, integrating $|e^{i\xi\cdot s}-1|^2|s|^{-d-1}$, and scaling/rotating $s$; that integral is finite at zero and infinity. Smooth chart maps on the compact supports are bilipschitz with bounded Jacobians, proving norm equivalence. Density then extends it to $H^{1/2}$. The local liftings, patched with the partition, give the global right inverse.

An $H^1$ field with zero trace has an $H^1$ zero extension in each flattened half-space: integration by parts has no boundary jump. Translate this extension a small distance toward the interior and mollify, then patch the charts and interior part. Translation continuity (Plancherel) and smooth mollification give convergence in $H^1$ by compactly supported interior smooth functions. This proves that the trace kernel is $H^1_0$. The same integration by parts shows that a glued field with matching traces has no derivative jump and belongs to $H^1$ locally.

For compactly supported vector fields, integration by parts gives

$$2\int|e(v)|^2=\int|\nabla v|^2+\int|\operatorname{div}v|^2.$$

Approximation extends this identity to $H^1_0$. Write $e=e_0+(\operatorname{tr}e)I/3$; pointwise elastic energy is $2\mu|e_0|^2+(\lambda+2\mu/3)|\operatorname{tr}e|^2$. Its positive continuous coefficients on the compact closure give a uniform

positive multiple of $|e|^2$. Poincaré follows for a zero extension in a surrounding cube by integrating each component from one face and applying one-dimensional Cauchy–Schwarz. Thus the energy is coercive on $H^1_0$. Lift the boundary datum, then solve for its zero-trace correction using Lemma A.1. Uniqueness follows from the same coercivity. The weak definition of traction is independent of the lifting by the weak equation, and is bounded by the trace right-inverse bound. This also proves complexification by splitting into real and imaginary parts.

For smooth extension, use boundary collars and their normal coordinate $t$, with the outside at $t > 0$. If the normal jets are $f_k(y)$ at $t = 0$, the outside series $\sum_{k\geq 0} \chi(t/\epsilon_k) f_k(y) t^k/k!$ has those jets. Take $\chi = 1$ near zero; choose $\epsilon_k \downarrow 0$ so that the $k$th term has all derivatives of orders at most $k-1$ bounded by $2^{-k}$ on each compact chart, which is possible because these derivative bounds contain positive powers of $\epsilon_k$. In every fixed derivative order the tail converges uniformly. Its boundary derivatives are precisely the prescribed jets; matching the inside function gives a smooth function. A finite partition of unity patches the collars and the inside region. This is also a direct proof of the extension when smoothness up to the boundary is initially defined by continuous one-sided derivatives. □

## A.2 Qualitative elliptic regularity

**Lemma A.4** (Local regularity for the operators used here)**.** *Let $P$ be a smooth matrix differential operator of order $m = 1$ or $2$ whose principal symbol is invertible for real $\xi \neq 0$ on a neighborhood of a compact set. If $u$ is locally $L^2$ and $Pu = f$ is locally $H^s$ for a nonnegative integer $s$, then $u$ is locally $H^{s+m}$. Smooth $f$ gives smooth $u$. The same smoothness conclusion holds for a distributional solution of $Pu = 0$. This applies to the conjugated Lamé operator and to the planar Cauchy–Riemann systems.*

*Proof.* Here is a local matrix construction requiring no unproved full symbolic calculus. A compact-$x$ symbol $q$ of order $r$ has

$$|\widehat{q}(\eta - \xi, \xi)| \leq C_N \langle \eta - \xi \rangle^{-N} \langle \xi \rangle^r$$

for every $N$, by integration by parts in $x$. Define $Qv(x) = (2\pi)^{-d/2} \int e^{ix\cdot\xi} q(x,\xi) \widehat{v}(\xi)\, \mathrm{d}\xi$ initially on Schwartz functions. Its weighted Fourier kernel is integrably bounded in both variables, since $\langle \eta \rangle^{t-r} \langle \xi \rangle^{r-t} \leq C \langle \eta - \xi \rangle^{|t-r|}$. Specifically, choose $N > d + |t-r| + 1$. If $K$ is this weighted kernel,

$$\Big| \int K(\eta,\xi) v(\xi)\, \mathrm{d}\xi \Big|^2 \leq \Big( \int |K(\eta,\xi)|\, \mathrm{d}\xi \Big) \int |K(\eta,\xi)| |v(\xi)|^2\, \mathrm{d}\xi.$$

Integrating in $\eta$ and applying integrable Fubini proves $Q : H^t \to H^{t-r}$. Extension by density and Hermitian duality prove the corresponding gain for its adjoint $Q^*$, for every real $t$.

Work in one small neighborhood and let $a^*_m(x,\xi)$ be the principal symbol of the formal Hermitian adjoint $P^*$. Its inverse and derivatives are homogeneous of degree $-m$ for $|\xi| > 0$, with smooth bounded angular parts on the compact neighborhood. Choose a compact cutoff $\chi$ in that neighborhood, equal to one on a smaller one, and a frequency cutoff $\chi_\infty$ equal to one for $|\xi| \geq 2$ and zero for $|\xi| \leq 1$. Take $q = \chi(x)\chi_\infty(\xi)(a^*_m)^{-1}(x,\xi)$, of order $-m$. Applying the differential operator $P^*$ on the left differentiates $e^{ix\cdot\xi} q$ by the finite Leibniz rule. The term with all $m$ derivatives on the exponential is $\chi\chi_\infty I$; every other term has order at most $-1$. Therefore, exactly,

$$P^*Q = \chi I + R, \qquad Q^*P = \chi I + R^*,$$

where $R$ has a compact-$x$ symbol of order $-1$; the low-frequency $\chi(\chi_\infty - 1)I$ is included in $R$. The identities first hold on tests and extend to compactly supported distributions by duality and the Sobolev bounds.

Choose $\psi = 1$ near $\operatorname{supp}\chi$. Applying the second identity to $\psi u$ gives

$$\chi u = Q^*(\psi f) - R^*(\psi u).$$

There is no commutator error here: $Q = \chi Q_0$, so $Q^* = Q_0^*\chi$, and $\chi[P,\psi]u = 0$ because $\psi$ is constant near $\operatorname{supp}\chi$. The first term gains $m$ derivatives and the second one gains one. Starting with local $H^0$, and shrinking cutoffs at each step, this gains one derivative repeatedly until $H^{s+m}$ is reached. If $f$ is smooth, all orders follow. Sobolev embedding proves smoothness. A compactly supported distribution has finite order by continuity on the test space; testing it on $\psi e^{-ix\cdot\xi}$ bounds its transform by a polynomial. It consequently belongs to $H^{-N}$ for some $N$. The same iteration starts there for the homogeneous distributional assertion.

For the Lamé principal symbol, with the positive sign convention, it is $m|\xi|^2 I + k\xi\xi^T$; its eigenvalues are $m|\xi|^2$ and $\ell|\xi|^2$, both positive. Conjugation changes lower orders only. For a planar system its symbol is a nonzero scalar multiple of $(i\xi_1 - \xi_2)I$. These symbols meet precisely the hypothesis used above. □

**Lemma A.5** (Quantitative local elastic estimate)**.** *On $B_2 \subset \mathbb{R}^3$ let $P = -\operatorname{div}(C\nabla) + \sum B_a\partial_a + Z$, where $C$ is the real isotropic elastic stiffness tensor with $\mu \geq c_0 > 0$ and $\lambda + 2\mu/3 \geq c_0 > 0$. Suppose all principal and lower coefficients and their first derivatives are bounded by $M$. The lower coefficients may be complex matrices. For locally $L^2$ distributional solutions $Pu = F$, with $F \in L^2(B_2)$,*

$$\|u\|_{H^2(B_1)} \leq C(c_0, M)(\|u\|_{L^2(B_2)} + \|F\|_{L^2(B_2)}).$$

*For smooth coefficients this estimate applies without a prior $H^1$ assumption, by Lemma A.4.*

*Proof.* Denote the principal energy by $a(v,w)$. The strain decomposition and Korn identity in Lemma A.3 give $a(v,v) \geq c\|\nabla v\|_2^2$ for compactly supported $v$, uniformly in the stated family. For a real cutoff $\phi$,

$$\operatorname{Re} a(u,\phi^2 u) = a(\phi u, \phi u) - \int [\lambda|u\cdot\nabla\phi|^2 + 2\mu|\operatorname{sym}(u\otimes\nabla\phi)|^2].$$

This is the direct product expansion using the real major-symmetric stiffness tensor; the last integrand may have either sign and its absolute integral is at most $C_M\|u\|_2^2$. The lower terms tested against $\phi^2 u$ are bounded by $C_M\|\phi\nabla u\|_2\|\phi u\|_2 + C_M\|u\|_2^2$. Use $\phi\nabla u = \nabla(\phi u) - u\otimes\nabla\phi$ and Young's inequality to absorb the gradient term. The forcing is bounded by $\|F\|_2\|u\|_2$. With $\phi = 1$ on an inner ball this gives a uniform $H^1$ bound on $B_{7/4}$ by $\|F\|_2 + \|u\|_2$ on $B_2$.

Put $w = \partial_s u$. In distributions,

$$Pw = \partial_s F + \operatorname{div}((\partial_s C)\nabla u) - \sum_a (\partial_s B_a)\partial_a u - (\partial_s Z)u.$$

Test with $\phi^2 w$, now with $\phi$ supported in $B_{7/4}$ and equal to one on $B_1$. Integrate the first two right-side terms by parts. Their absolute value is at most

$$C(\|F\|_2 + \|\nabla u\|_2)(\|\nabla(\phi w)\|_2 + \|w\|_2).$$

The remaining terms have bound $C(\|\nabla u\|_2 + \|u\|_2)\|w\|_2$. The preceding energy argument for $w$, and Young's inequality, absorb the gradient of $\phi w$. The first-step bound controls $\|w\|_2$ and the other $u$ terms on this support. Summing over $s$ proves the displayed $H^2$ estimate with the stated uniform constant. Lemma A.4 first gives $u \in H^2_{\mathrm{loc}}$, so all tests are legitimate by smooth approximation; $\partial_s F$ acts on $H^1_0$ and needs no derivative bound for $F$. □

### A.3 Planar analytic facts and propagation

**Lemma A.6** (Planar Cauchy facts)**.** *With $\partial_{\bar z} = (\partial_{y_1} + i\partial_{y_2})/2$, $\partial_{\bar z}(1/(\pi z)) = \delta_0$. Its convolution with bounded data supported in a disk of radius $\delta$ has $L^\infty$ norm at most $C\delta\|f\|_\infty$. Smooth Cauchy–Riemann functions obey the identity theorem; holomorphic functions on the compact null conic are constant.*

*Proof.* Apply Green's formula on a punctured disk to $\varphi/z$. The outer boundary is zero for a compact test, and the inner boundary has the opposite orientation; its limit is $-2\pi i\varphi(0)$. Since $\int \partial_{\bar z} v \,\mathrm{d}y = (2i)^{-1}\int_{\partial D} v\,\mathrm{d}z$, the distributional identity follows with the stated sign and factor. Convolution is well-defined because $|z|^{-1}$ is locally integrable. The integral over the support disk is bounded by $C\delta$ uniformly in $z$: use a disk of radius $3\delta$ about $z$ when it is close, and the distance bound otherwise. Integrable Fubini justifies differentiation in distributions. For smooth data, the convolution is smooth locally by Lemma A.4.

For a smooth Cauchy–Riemann function the same punctured-disk calculation gives Cauchy's integral formula. Expanding $(w-z)^{-1}$ by its uniformly convergent geometric series on a smaller concentric disk gives a power series there. Its first nonzero coefficient isolates each zero unless all coefficients vanish. Overlapping disks therefore prove the identity theorem. Cauchy's formula also gives the mean-value identity. At a point of maximum modulus on a compact connected complex curve, equality in that mean identity forces local constancy (the values on the circle have the same phase and modulus); the identity theorem gives global constancy. The conic is compact and connected by its explicit parametrization from $\mathbb{P}^1$, the continuous quotient of the compact connected unit sphere in $\mathbb{C}^2$. Thus this conclusion applies to it. □

**Lemma A.7** (Elementary continuation along paths)**.** *If a smooth finite-dimensional vector $v$ satisfies a homogeneous smooth linear first-order system $\partial_s v = A_s(x)v$ on an open connected set, and vanishes on a nonempty open subset, then it vanishes everywhere.*

*Proof.* The points reachable from a fixed point by polygonal paths inside the set form a nonempty subset that is both relatively open and relatively closed: a small ball around any point supplies the last straight segment. Thus every point is reachable. On each compact path segment the chain rule gives $\dot v = A(t)v$ with bounded continuous coefficients. From zero initial data, $|v(t)| \le C\int_0^t |v(s)|\,\mathrm{d}s$. Iterating $N$ times bounds it by $\sup|v|\,C^N t^N/N!$, which tends to zero. A finite concatenation propagates the zero value along the path. This proves the assertion without an additional uniqueness input for ordinary differential equations. □

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